

Suei-Wen Chen

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EDUCATION

ETH Zürich *Master in Data Science* Sep 2024 - Present

Specialization: Finance and Insurance

New York University Abu Dhabi *Bachelor of Science* Aug 2019 - May 2023

Major: Mathematics, Minor: Computer Science, Cumulative GPA: 4.0/4.0 (summa cum laude)

WORK EXPERIENCE

Research Assistant | *Division of Science, New York University Abu Dhabi* Sep 2023 - Aug 2024

Worked on the mathematical foundations of machine learning under Dr. Pierre Youssef, including reinforcement learning, strategic games, language models, multi-modal data classification and differential privacy

Data Science Intern | *Swiftbox Delivery Service LLC, UAE* Mar 2022 - Feb 2023, Jul 2023 - Aug 2023

Produced data-driven and mathematical solutions to optimize last-mile logistics, including a geocoding pipeline, various prediction algorithms, and mathematical optimization for vehicle routing problems

Intern | *Arab Chamber of Commerce Taiwan, Taiwan* Jun 2020 - Aug 2020

Produced commercial clips for Taiwanese businesses targeting Arab countries

Teacher & Tutor | *Julia's English Classroom, Taiwan* Feb 2019 - Present

Give English classes and design teaching materials for various levels (from elementary school to TOEFL/IELTS)

RESEARCH PROJECTS

Finite-Sample Analysis of the Monte Carlo Exploring Starts Algorithm for Reinforcement Learning | Jan 2024-Aug 2024

Established convergence rate of the Monte Carlo Exploring Starts algorithm for a general set of environments

How Bad is Training on Synthetic Data? A Statistical Analysis of Language Model Collapse | Nov 2023 - Apr 2024

Published in *Conference on Language Modeling (COLM)* 2024

Quantified how training language models on generated data reduces output diversity (the *model collapse* phenomenon)

Markov Chains, Mixing Times and Functional Inequalities | *Undergraduate Thesis* Mar 2022 - May 2023

Studied the mixing properties of Markov chains and their relation to functional inequalities, proposed a decomposition result for establishing functional inequalities (*A Fuzzy Decomposition Method to Establish Functional Inequalities*)

PROJECTS

Column generation for mixed integer programming | Jul 2023 – Aug 2023

Combined graph-reduction algorithm and column generation for solving a modified vehicle routing problem

Reinforcement learning on vehicle routing problems | Dec 2022 - Feb 2023

Combined multi-arm bandit algorithm and local search methods for stochastic vehicle routing problem

Effects of colonialism on sentiments towards Queen Elizabeth II | Oct 2022 – Dec 2022

Emotion analysis on Tweets to compare the difference in sentiments towards the queen's death between former British colonies and other countries

Feature engineering for unbalanced class classification of delivery data |

Sep 2022 - Oct 2022

Process features for classification with unbalanced class using gradient boosting classifier and data augmentation

Smart addressing system |

Apr 2022 – Aug 2022

Named-entity recognition for addresses by combining deep learning (NLP) and graph-based methods; outperformed several commercially available geocoding services

Towards self-driving car |

Oct 2021 – Dec 2021

Application of the YOLOv3 algorithm on real-time object detection in videos for self-driving car use case

Debunking the myth regarding Ghost Month and drowning |

Oct 2021 – Dec 2021

Data analysis on drowning in relation to the Ghost Month, a traditional belief in East Asia

VOLUNTEER AND LEADERSHIP EXPERIENCE**Vice Director | *Global Devotion Association, Taiwan***

Jun 2017 – Aug 2023

Organized cultural exchange programs between Taiwan and the US, summer camps, volunteering and community outreach in aboriginal tribes and hospitals; promoted student initiatives and leadership

LANGAUGES AND SKILLS**Programming Langauges/Skills |**

- Langauges: Python, Julia, C++, Java, SageMath
- Large language models (LLMs), Natural language processing (NLP), Reinforcement learning (RL)
- Data science and machine learning: pandas, numpy, scipy, scikit-learn, PyTorch
- Data visualization: matplotlib, seaborn, plotly
- Database & Big data systems: MySQL, PostgreSQL, Neo4J, MongoDB, MapReduce
- Geospatial data analysis and visualization (OpenStreetMap)
- Probability theory, Mathematical optimization, Cryptography

Langauges |

Mandarin (native), English (proficient), Arabic (intermediate), German (beginner)